# Zain Dana Harper

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**Graphics and Rendering | Developer Tools | Languages**

## Summary

Self-directed graphics and tools programmer moving from eleven years of arboriculture; the career change began June 2026, with no professional programming role yet. Builds renderers, shader presets and compilers that check each fast path against a reference.

## Technical Skills

C++23, Rust, Python, HLSL, WGSL, TypeScript; D3D11, D3D12, Vulkan, WebGPU; CPU reference rendering and GPU parity checks; color science (Oklab, CAM16, ICC profiles); compilers, type inference, language servers; CMake, GitHub Actions, pytest.

## Self-Directed Projects

About 13 years self-directed | Career transition began June 2026

- Built a C++23 renderer (raw-native, now in my Superframe engine) with a CPU reference beside D3D12 and WebGPU backends; release 0.8.0 passed 60 of 60 tests with MSVC and GCC, and its default renders were byte-identical across Windows, D3D12, Linux and WebAssembly builds.
- Merged and not yet released: a Vulkan 1.3 backend whose 600 presented frames match the offscreen and D3D12 WARP renders hash for hash, radiance cascades measured against the path trace, and ray queries.
- Released Elder ENB, a lighting and post-processing preset for Skyrim Special Edition with 38 atmosphere presets: 1,004,693 downloads and 2,418 endorsements on Nexus Mods, read November 13, 2026.
- Developed BuildLang, a language with typed capability effects and C and shader output; 1.14.0 is on crates.io. WebAssembly and x86-64 native (first slice) backends are merged, not yet released.
- Brought Argonaut's 1990s BRender engine to 64-bit Windows and Linux in Brender Archival, and built Calibrate Pro (ICC profiles, 3D LUTs, DDC/CI) and Build Color (Oklab, CAM16, Delta E), both on PyPI.

### Freelance Technical Writer | Independent projects | Began 2017

- Wrote implementation guides, API documentation, and release notes for developers and customers.

## Employment

### Arboriculture and Field Operations | Legendary Tree

Seattle area | April 25, 2015 to June 2, 2026

- Turned site assessments into estimates, proposals, and workable job scopes; coordinated clients, crews, and vendors so teams understood access constraints, timing, and hazards.

### Technical Networking Support | Stream/Convergys | 2014 to 2015

Xbox/Microsoft product contract | Wilsonville, Oregon

- Resolved Xbox connectivity problems across TCP/IP, DNS, NAT, firewalls, and routers; documented troubleshooting steps for consistent support handoffs.

## Open-Source Contributions: 38 Merged in 31 Repositories

- TOMLKit #549: preserved sibling keys in dotted-key replacement, preventing silent configuration loss. Merged.
- Datasette #2815: corrected malformed composite-key requests from HTTP 500 to HTTP 400, with tests. Merged.

## Certifications and Education

Security and cloud certifications from Qualys, Microsoft Applied Skills and Google Cloud, none specific to graphics; all 42 credentials, with verification links: harperz9.github.io/published-work.html

High School Diploma | Wilsonville High School, Oregon | 2013